No.	Status	Date
LNDE-RPT-20140902-024-DS-4-RA-102	Issued	2014-09-19

UT Thickness Data Sheet - 4-RA-102

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurements for line 4-RA-102 located within the ATU-100. The scope of work consists of a pre-change rich-amine survey to establish baseline thickness data prior to the planned MOC, as detailed in LNDE-RPT-20140902-024-R0.

Marcus Chen, acting as the lead UT technician and SNT-TC-1A UT Level II for Larkhall NDE Services Inc., is responsible for the execution of the measurements and the recording of the thickness data.

Technical Basis

All thickness measurements were performed in accordance with QSCP-NDT-001 Rev 2. This procedure incorporates specific requirements for the CUI program, including the management of insulation removal and reinstatement and the use of inspection ports.

The ultrasonic testing was conducted following the guidelines set forth in ASTM E797 to ensure accurate wall thickness determination.

Component Specifications

The subject line, 4-RA-102, is constructed from A106 Gr.B with a nominal outer diameter of 114.3 mm and a nominal wall thickness of 8.56 mm. The design parameters for this service are 862 kPa(g) and 82 °C.

The minimum allowable thickness (t-min) for this line is 3.81 mm. Any reading recorded below this value is subject to an NCR per site integrity rules.

Inspection Interval and Context

The rich amine service in the ATU-100 is subject to a 3-year thickness measurement interval. This survey follows the previous rich-amine readings recorded in 2011-02.

The current data is integrated into the broader survey documented in LNDE-RPT-20140902-024-R0 to determine the short-term and long-term corrosion rates for remaining life calculations.

Thickness readings, measured 2014-09-19

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	2011-02-18	8.32	8.07	0.024	0.070	>50	2017-09-19	
4-RA-101-02	8.56	3.81	2011-02-18	8.20	8.02	0.028	0.050	>50	2017-09-19	
4-RA-101-03	8.56	3.81	2011-02-18	7.98	7.88	0.039	0.028	>50	2017-09-19	
4-RA-101-04	8.56	3.81	2011-02-18	8.21	8.18	0.024	0.008	>50	2017-09-19	
4-RA-101-05	8.56	3.81	2011-02-18	8.16	7.98	0.033	0.050	>50	2017-09-19	
4-RA-101-06	8.56	3.81	—	—	7.91	—	—	—	2017-09-19	
4-RA-101-07	8.56	3.81	2011-02-18	8.30	8.22	0.021	0.022	>50	2017-09-19	
4-RA-101-08	8.56	3.81	2011-02-18	8.02	7.71	0.052	0.086	45.1	2017-09-19	
4-RA-101-09	8.56	3.81	2011-02-18	8.32	8.27	0.016	0.014	>50	2017-09-19	
4-RA-101-10	8.56	3.81	2011-02-18	8.28	8.21	0.025	0.020	>50	2017-09-19	
4-RA-101-11	8.56	3.81	2011-02-18	8.04	7.88	0.043	0.045	>50	2017-09-19	

Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-102-01	8.56	3.81	2011-02-18	8.33	8.25	0.016	0.022	>50	2017-09-19	
4-RA-102-02	8.56	3.81	2011-02-18	7.99	7.89	0.037	0.028	>50	2017-09-19	
4-RA-102-03	8.56	3.81	2011-02-18	8.04	7.95	0.033	0.025	>50	2017-09-19	
4-RA-102-04	8.56	3.81	2011-02-18	7.96	7.93	0.033	0.008	>50	2017-09-19	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-102-05	8.56	3.81	2011-02-18	8.07	7.84	0.038	0.064	>50	2017-09-19	
4-RA-102-06	8.56	3.81	2011-02-18	8.01	7.95	0.030	0.017	>50	2017-09-19	
4-RA-102-07	8.56	3.81	2011-02-18	8.10	7.98	0.032	0.033	>50	2017-09-19	
4-RA-102-08	8.56	3.81	2011-02-18	7.88	7.58	0.049	0.084	45.0	2017-09-19	
4-RA-102-09	8.56	3.81	2011-02-18	7.98	7.84	0.038	0.039	>50	2017-09-19	
4-RA-102-11	8.56	3.81	2011-02-18	8.19	8.12	0.020	0.020	>50	2017-09-19	

Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-01	10.97	4.57	2011-02-18	10.47	10.62	0.021	-0.042	>50	2017-09-19	
6-RA-103-02	10.97	4.57	2011-02-18	10.23	10.06	0.056	0.047	>50	2017-09-19	
6-RA-103-03	10.97	4.57	2011-02-18	10.42	10.39	0.032	0.008	>50	2017-09-19	
6-RA-103-04	10.97	4.57	2011-02-18	9.97	9.79	0.064	0.050	>50	2017-09-19	
6-RA-103-05	10.97	4.57	2011-02-18	10.10	9.81	0.062	0.081	>50	2017-09-19	
6-RA-103-06	10.97	4.57	2011-02-18	10.56	10.43	0.031	0.036	>50	2017-09-19	
6-RA-103-07	10.97	4.57	2011-02-18	10.55	10.37	0.033	0.050	>50	2017-09-19	
6-RA-103-08	10.97	4.57	2011-02-18	10.44	10.49	0.026	-0.014	>50	2017-09-19	
6-RA-103-09	10.97	4.57	2011-02-18	10.56	10.39	0.033	0.047	>50	2017-09-19	
6-RA-103-10	10.97	4.57	2011-02-18	10.05	9.93	0.059	0.033	>50	2017-09-19	
6-RA-103-11	10.97	4.57	—	—	10.10	—	—	—	2017-09-19	
6-RA-103-13	10.97	4.57	2011-02-18	10.45	10.38	0.020	0.020	>50	2017-09-19	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-01	8.56	3.81	2011-02-18	7.95	7.84	0.036	0.031	>50	2017-09-19	
4-RA-104-02	8.56	3.81	2011-02-18	8.14	8.09	0.025	0.014	>50	2017-09-19	
4-RA-104-03	8.56	3.81	2011-02-18	8.56	8.46	0.032	0.028	>50	2017-09-19	
4-RA-104-04	8.56	3.81	2011-02-18	8.60	8.49	0.007	0.031	>50	2017-09-19	
4-RA-104-05	8.56	3.81	2011-02-18	8.57	8.37	0.015	0.056	>50	2017-09-19	
4-RA-104-06	8.56	3.81	2011-02-18	7.98	7.91	0.037	0.020	>50	2017-09-19	
4-RA-104-07	8.56	3.81	2011-02-18	7.78	7.61	0.053	0.047	>50	2017-09-19	
4-RA-104-08	8.56	3.81	2011-02-18	8.18	8.00	0.050	0.050	>50	2017-09-19	

Sign-off	Name	Function	Date	Signature
Prepared	Marcus Chen	lead UT technician, SNT-TC-1A UT Level II	2014-09-19	_____